MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
Fourth SEMESTER
Discipline Specific Core Course (DSC) - MG4DSCAQC202 -
REPRODUCTIVE PHYSIOLOGY, ENDOCRINOLOGY AND INDUCED BREEDING TECHNIQUES
(2024 ADMISSION ONWARDS)
Duration: 1 hour 30 minutes
Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Very Short Answer questions
Answer all questions
Each question carries 1 marks

1. Mention one advantage of sex reversal in fishes

1. Production of Monosex Populations 2. Faster Growth Rate, 3. Prevention of Unwanted Breeding, 4. Better Feed Utilization, 5. Uniform Size and Quality, 6. Increased Yield and Profit, 7. Efficient Use of Culture Systems, 8. Control Over Breeding Programs (mention any one)

2. Mention the sexual dimorphism between male and female *Amia calva*

Male

Possesses a distinct black spot (ocellus) at the base of the caudal (tail) fin, usually surrounded by an orange or yellow ring.

Body may appear slightly more slender.

During breeding season, males guard the nest and eggs.

Female

Lacks the prominent black spot at the base of the caudal fin

Generally broader or fuller body, especially when carrying eggs.

Does not exhibit nest-guarding behavior.

3. What is meany by genetic sex determination?

Genetic sex determination (GSD) in fishes refers to a system in which the sex of an individual is determined by its genetic makeup (chromosomes or genes) at the time of fertilization.

Sex is fixed by sex chromosomes or specific genes inherited from parents.

It is similar to mechanisms seen in other animals, where combinations like XX/XY or ZZ/ZW determine sex.

Once determined, the sex generally does not change due to environmental factors (though some exceptions exist in fishes).

Examples:

XX/XY system:

XX → Female

XY → Male

ZZ/ZW system:

ZZ → Male

ZW → Female

4. What is gonadotropin releasing inhibitory factor?

Gonadotropin Releasing Inhibitory Factor (GRIF) is a neuropeptide hormone that suppresses reproductive activity. It is a hormone produced in the hypothalamus that inhibits the release of gonadotropins from the pituitary gland. Eg. Dopamine

5. Mention the hormones involved in migration and concentration of pigments in epidermal chromatophores and retinal cells in crustaceans

In crustaceans, the migration and concentration of pigments in epidermal chromatophores and retinal cells are controlled by neurohormones secreted from the eyestalk (X-organ–sinus gland complex).

Hormones involved:

1. Pigment Concentrating Hormone (PCH)
    1. Causes concentration of pigments in chromatophores
    2. Leads to a lighter body colour
2. Pigment Dispersing Hormone (PDH)

    1. Causes dispersion of pigments in chromatophores
    2. Results in a darker body colour
3. Distal Retinal Pigment Hormone (DRPH)

    1. Controls movement of retinal pigments in the eye
    2. Helps in light and dark adaptation of vision

6. Outline the function of GTH 1 and 2

Gonadotropins (GTH) in fishes are two pituitary hormones involved in reproduction:

GTH I (Gonadotropin I)

Promotes early gonadal development, Stimulates gametogenesis , In females: supports vitellogenesis (yolk formation in oocytes), In males: aids spermatogenesis

GTH II (Gonadotropin II)

Responsible for final maturation of gametes, In females: induces ovulation, In males: stimulates spermiation, Triggers spawning behavior

7. Name the hormone present in eye stalk that inhibits maturity and spaawning in crustaceans

Gonad Inhibiting Hormone (GIH).

GIH is secreted from the eyestalk (X-organ–sinus gland complex) of crustaceans.

It inhibits gonadal maturation and spawning by suppressing reproductive activity.

8. Name any two artificial hormones used in induced breeding of fin fishes

1. Human Chorionic Gonadotropin (HCG)

2. Gonadotropin-Releasing Hormone analogues (GnRHa), SGnRHa

3. Commercial hormone preparations

Ovaprim (GnRHa + dopamine antagonist), Ovatide, WOVA-FH

9. What is the method of preservation of biological material at very low temperature in liquid nitrogen?

Cryopreservation.

Cryopreservation is the technique of preserving biological materials (such as gametes, embryos, tissues, or cells) at very low temperatures, typically in liquid nitrogen (−196 °C). At this temperature, all metabolic activities are completely halted, allowing long-term storage without damage.

10. What is invagination?

Invagination in embryology is the infolding of a part of a cell layer into the interior of the embryo, It is a major movement during gastrulation., A region of the blastula folds inward, forming a cavity. This process leads to the formation of the archenteron, Helps in the development of germ layers.

Part B

Short Answer Questions
Answer any **5** questions
Each question carries **6** marks

11. Describe sexual dimorphism in crabs

Abdomen shape: Male – narrow, triangular, Female – broad, rounded

Pleopods : Male – modified (copulatory), Female – well developed (egg carrying)

Chelipeds (claws): Male – larger, stronger, Female – smaller

Body size: Male – usually larger, Female – smaller (but broader abdomen)

Gonopores position: Male – on base of 5th walking legs, Female – on base of 3rd walking legs

Abdominal segments: Male – fewer free segments, Female – more expanded for brooding

12. Explain male reproductive system in prawns with suitable diagram

Paired testes – elongated, lobulated , Location – dorsal side of cephalothorax , Colour – whitish/cream, Vasa deferentia – paired ducts from testes, Regions – proximal, medial, distal parts, Function – transport and maturation of sperm , Seminal vesicle – terminal enlarged part of vas deferens, Function – storage of spermatophores, Spermatophores – packeted sperm, Gonopores – openings at base of 5th walking legs, Androgenic gland – associated with vas deferens , Function – controls male characteristics, Appendix masculina – present on 2nd pleopod Function – transfer of spermatophores

Draw diagram

13. Describe oviparity, viviparity and ovoviviparity in fishes

Oviparity: Egg laying, Fertilization – external (mostly), Development – outside body , Large number of eggs, Less parental care

Viviparity: Live bearing, Fertilization – internal, Embryo develops inside female, Nourishment from mother, Fewer young, more protection

Ovoviviparity: Eggs retained inside female, Fertilization – internal, Development inside egg within mother, Nourishment from yolk (no direct maternal supply), Young released as live

14. Discuss the position and functions of thyroid gland, ultimobranchail gland, pineal organ, chromaffin tissues and pancreatic islets in teleosts

Thyroid gland

Position – Diffuse follicles around ventral aorta, near gill region (pharyngeal area)

Function – Secretes thyroid hormones (T_3, T_4); regulates metabolism, growth, metamorphosis, osmoregulation

Ultimobranchial gland

Position – Near oesophagus/posterior pharyngeal region

Function – Secretes calcitonin; regulates calcium metabolism (lowers blood Ca^{2+})

Pineal organ

Position – Dorsal side of brain (epithalamus)

Function – Secretes melatonin; controls biological rhythms, photoperiodism, reproduction

Chromaffin tissues

Position – Associated with kidney (head kidney/interrenal tissue)

Function – Secretes adrenaline & noradrenaline; involved in stress response, cardiovascular regulation

Pancreatic islets (Islets of Langerhans)

Position – Scattered in pancreas/hepatopancreas

Function – Secretes insulin, glucagon; regulates blood glucose metabolism

15. Explain the position and functions of the Y organ and mandibular organs in crustacea

Y-organ

Position – Anterior region of cephalothorax, near base of maxillipeds, Hormone – Ecdysone (moulting hormone), Function – Initiates moulting (ecdysis), growth Control – Inhibited by MIH from eyestalk

Mandibular organ

Position – At base of mandibles, Hormone – Methyl farnesoate, Function – Regulates reproduction, gonadal maturation, Role – Influences growth and development

16. Describe the advantages of producing monosex population in aquaculture

Faster growth – one sex grows faster (e.g., males in many fishes), Uniform size – reduced size variation, Higher yield – improved production, Prevents unwanted breeding – avoids overcrowding, Better feed conversion – energy used for growth, not reproduction, Improved management – easier culture practices, Better survival

rate – less competition, Enhanced market value – uniform and desirable size , Efficient use of space – suitable for high-density culture

17. Describe the procedure of hypophysation

Hypophysation (induced breeding using pituitary extract)

Selection of brooders – healthy, mature male & female, Collection of pituitary gland – from donor fish, Preparation of extract – pituitary homogenized in distilled water/saline, Dose determination – based on body weight & species Injection – intramuscular/intraperitoneal, Two doses (female): Preliminary (priming) doseand Final (resolving) dose, Single dose (male) – at time of female's final dose, Latency period – time between injection & spawning, Spawning – natural or artificial (stripping), Fertilization – mixing eggs & milt, Incubation – hatching of eggs under controlled conditions

18. Describe the gonadal level of control in induced maturation and spawning

Target organ – Gonads (ovary & testis)

Steroid hormones – Estrogens (e.g., estradiol) – vitellogenesis (yolk formation), Androgens – spermatogenesis, Progestogens (maturation-inducing hormone, MIH) – final maturation

Ovary – Oocyte growth, Vitellogenesis, Final oocyte maturation, Ovulation

Testis - Spermatogenesis, Sperm maturation, Spermiation

Maturation Inducing Hormone (MIH) – Triggers germinal vesicle breakdown (GVBD), Leads to final oocyte maturation

Feedback control – Steroids regulate pituitary & hypothalamus

Part C

Essay Questions

Answer any **1** questions

Each question carries **10** marks

19. Explain the maturity stages of male and female in teleosts with diagrams

Female maturity stages

Stage I – Immature : Ovary – small, thin, transparent, Oocytes – oogonia, primary oocytes, No yolk , Thin thread-like ovary

Stage II – Early maturing: Ovary – slightly enlarged, Oocytes – previtellogenic , Beginning of cytoplasmic growth, Slightly enlarged ovary

Stage III – Maturing: Ovary – enlarged, yellowish, Oocytes – vitellogenic (yolk deposition), Yolk vesicles present, Enlarged ovary with yolk

Stage IV – Mature: Ovary – large, fills body cavity, Oocytes – fully yolked , Eggs clearly visible, Large ovary full of eggs

Stage V – Ripe/Spawning: Ovary – turgid, soft, Eggs – hydrated, ready to release , Ovulation occurs, Ovary releasing eggs

Stage VI – Spent: Ovary – shrunken, flaccid, Residual eggs present, Post-spawning condition, Shrunken ovary

Male maturity stages

Stage I – Immature: Testis – small, thread-like, No spermatozoa , Thin testis

Stage II – Early maturing: Testis – slightly enlarged, Spermatogonia present

Slightly enlarged testis

Stage III – Maturing: Testis – enlarged, whitish, Spermatocytes, spermatids, Enlarged testis

Stage IV – Mature: Testis – large, milky white, Spermatozoa abundant , Full testis

Stage V – Ripe/Running: Testis – soft, Milt oozes on pressure , Milt release

Stage VI – Spent, Testis – reduced, flaccid, Residual sperm, Shrunken tests

20. Summarise the different hormones produced by the pituitary and urophysis in fishes and explain their functions

Pituitary gland (Hypophysis)

Adenohypophysis (Anterior pituitary)

Growth Hormone (GH) : Growth, protein synthesis, Regulates metabolism, osmoregulation

Prolactin (PRL): Freshwater adaptation, Ion regulation, parental behaviour

Thyroid Stimulating Hormone (TSH) : Stimulates thyroid gland, Controls metabolism, metamorphosis

Adrenocorticotropic Hormone (ACTH) : Stimulates interrenal tissue, Stress response (cortisol release)

Gonadotropins (GTH I & GTH II) : GTH I – gametogenesis, vitellogenesis , GTH II – maturation, ovulation, spermiation

Melanocyte Stimulating Hormone (MSH) : Pigment dispersion, Skin colour regulation

Neurohypophysis (Posterior pituitary)

Vasotocin (AVT) : Osmoregulation, Reproductive behaviour

Isotocin : Oviduct contraction, Spawning, parturition

Urophysis (Caudal neurosecretory system)

Urotensins (UI, UII) : Ion balance (Na^+, Cl^- regulation), Cardiovascular effects , Smooth muscle contraction

Urotensin III / related peptides : Osmoregulation, Stress adaptation

21. Describe anesthetics used in fish breeding and handling, their importance and methods of administering anesthetics

Common anaesthetics used

MS-222 (Tricaine methanesulfonate) : Most widely used, Rapid action, safe

Benzocaine : Cheap, effective, Requires alcohol for dissolution

2-Phenoxyethanol : No buffering needed, Stable in water

Clove oil (Eugenol) : Natural, low cost, Common in hatcheries

Quinaldine : Used in marine fishes

Importance / Uses: Reduces stress during handling, Prevents injury to fish & handler , Facilitates breeding operations

Stripping of eggs & milt , Artificial fertilization

Ease of transport , Accurate measurements (length, weight), Tagging, surgery, sampling, Improves survival rate, Minimizes metabolic activity

Methods of administration

Immersion (most common) : Fish placed in anaesthetic solution, Absorption through gills

Dip method : High concentration, short duration

Prolonged bath : Low concentration, long exposure

Irrigation method : Anaesthetic water passed over gills, Used during surgery

Stages of anaesthesia

Light sedation → loss of equilibrium → deep anaesthesia → recovery